

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected Graduation, 2027

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding entirely by automating member provisioning (GitHub API + Slack SDK) to grant full repo and workspace access in 30 seconds, cutting setup time 97% across 100+ members.
- Crafted a cron-scheduled event pipeline that cross-checks check-in records with enrollment and posts robust, per-event Slack reports—bad records never break the batch thanks to soft-fail, idempotent functions.
- Replaced paper sign-in sheets with a QR-driven check-in system backed by AWS S3 token caching, giving board members instant visibility into live participation data and cutting processing time by 80%.
- Enforced a 4-tier role hierarchy (prospect → member → board → e-board) at the database layer using Postgres row-level security, so every new API route and query inherits access rules automatically with no per-route auth boilerplate.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Refactored a relational query layer with a graph database (Neo4j) to model connected data natively; relationship-type indexing eliminated full-table scans and cut query latency 48%, making traversals 3× faster than equivalent SQL joins.
- Integrated Google and GitHub OAuth 2.0 so users authenticate without passwords; secured sessions with httpOnly JWT cookies and automatic token rotation to prevent credential exposure between requests.
- Built a GitHub Actions CI/CD pipeline running integration test suites inside a Dockerized environment; every pull request triggers automated validation against 85% coverage gates before merge is permitted.

PROJECTS

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery*

May 2025 – Nov. 2025

- Engineered a multi-client voice assistant that listens to natural conversation, infers what each speaker wants, and dispatches tasks across concurrent audio streams.
- Designed cross-platform desktop client (Electron + Next.js) with an embedded HTTP server for OAuth callback handling; real-time event propagation via IPC delivers sub-50ms latency.
- Architected an async ML inference pipeline with Redis/Celery to offload Whisper and Sentence-BERT from the request path, cutting voice endpoint response time to under 200ms.
- Routed processed results back to clients through a dedicated webhook server, sustaining sub-100ms queue pickup latency across 10 concurrent voice streams.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

- Won Interactive Media Track: Gesture-controlled wizard duel where full-body movements cast spells, block attacks, and trigger abilities against an AI opponent.
- Built full-body gesture pipeline using MediaPipe Pose to map 33 skeletal landmarks into game actions; optimized inference loop to sustain 60 FPS with sub-20ms input lag.
- Developed a custom 2D physics engine with collision detection to translate live skeletal coordinates into game state, maintaining <5px positional accuracy at 60 FPS.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Cypher, Golang, C++, Bash
Frameworks Spring Boot, FastAPI, Node.js, Angular, React, Next.js, Redis
Databases PostgreSQL, Neo4j, MongoDB
Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitHub Actions, CI/CD Pipelines